

.env .gitignore U product_listing_generator.ipynb U

product_listing_generator.ipynb > # Install: pip install datasets

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api_lab_conda (Python 3.11.15)

```
# Load dataset from HuggingFace
print("Loading product dataset...")
try:
    # Try loading the dataset
    dataset = load_dataset("ashraq/fashion-product-images-small", split="train[:10]") # First 100 samples
    print(f"✓ Loaded {len(dataset)} products")

    # Convert to pandas for easier manipulation
    products_df = pd.DataFrame(dataset)
    print(f"Dataset columns: {products_df.columns.tolist()}")

except Exception as e:
    print(f"⚠ Could not load HuggingFace dataset: {e}")
    print("Using local images instead...")

    # Alternative: Use local images
    # Create a products.json file with product information
    products_data = [
        {
            "id": 1,
            "name": "Wireless Headphones",
            "price": 79.99,
            "category": "Electronics",
            "image_path": "images/product1.jpg"
        },
        # Add more products...
    ]

    products_df = pd.DataFrame(products_data)

# Create images directory
images_dir = Path("product_images")
images_dir.mkdir(exist_ok=True)

print(f"\n✓ Dataset prepared!")
print(f"  Total products: {len(products_df)}")
```

[3]

✓ 29.7s

Python

```
/opt/miniconda3/envs/api_lab_conda/lib/python3.11/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please
from .autonotebook import tqdm as notebook_tqdm
Loading product dataset...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fa
Generating train split: 100%|██████████| 44072/44072 [00:00<00:00, 134530.70 examples/s]
✓ Loaded 10 products
Dataset columns: ['id', 'gender', 'masterCategory', 'subCategory', 'articleType', 'baseColour', 'season', 'year', 'usage',
✓ Dataset prepared!
Total products: 10
```

```
# Load dataset from HuggingFace
print("Loading product dataset...")
try:
    # Try loading the dataset
    dataset = load_dataset("ashraq/fashion-product-images-small", split="train[:50]") # First 100 samples
    print(f"✓ Loaded {len(dataset)} products")

    # Convert to pandas for easier manipulation
    products_df = pd.DataFrame(dataset)
    print(f"Dataset columns: {products_df.columns.tolist()}")

except Exception as e:
    print(f"⚠ Could not load HuggingFace dataset: {e}")
    print("Using local images instead...")

    # Alternative: Use local images
    # Create a products.json file with product information
    products_data = [
        {
            "id": 1,
            "name": "Wireless Headphones",
            "price": 79.99,
            "category": "Electronics",
            "image_path": "images/product1.jpg"
        },
        # Add more products...
    ]

    products_df = pd.DataFrame(products_data)

# Create images directory
images_dir = Path("product_images")
images_dir.mkdir(exist_ok=True)

print(f"\n✓ Dataset prepared!")
print(f"  Total products: {len(products_df)}")
```

[4] ✓ 1.3s Python

```
... Loading product dataset...
✓ Loaded 50 products
Dataset columns: ['id', 'gender', 'masterCategory', 'subCategory', 'articleType', 'baseColour', 'season', 'year', 'usage',

✓ Dataset prepared!
  Total products: 50
```

```
# Load dataset from HuggingFace
print("Loading product dataset...")
try:
    # Try loading the dataset
    dataset = load_dataset("ashraq/fashion-product-images-small", split="train[:100]") # First 100 samples
    print(f"✓ Loaded {len(dataset)} products")

    # Convert to pandas for easier manipulation
    products_df = pd.DataFrame(dataset)
    print(f"Dataset columns: {products_df.columns.tolist()}")

except Exception as e:
    print(f"⚠ Could not load HuggingFace dataset: {e}")
    print("Using local images instead...")

    # Alternative: Use local images
    # Create a products.json file with product information
    products_data = [
        {
            "id": 1,
            "name": "Wireless Headphones",
            "price": 79.99,
            "category": "Electronics",
            "image_path": "images/product1.jpg"
        },
        # Add more products...
    ]

    products_df = pd.DataFrame(products_data)

# Create images directory
images_dir = Path("product_images")
images_dir.mkdir(exist_ok=True)

print(f"\n✓ Dataset prepared!")
print(f"  Total products: {len(products_df)}")
```

[S] ✓ 1.2s Python

```
... Loading product dataset...
✓ Loaded 100 products
Dataset columns: ['id', 'gender', 'masterCategory', 'subCategory', 'articleType', 'baseColour', 'season', 'year', 'usage',

✓ Dataset prepared!
  Total products: 100
```

```
Format your response as JSON with the following structure:
{{
  "title": "Product title here",
  "description": "Full description here",
  "features": ["Feature 1", "Feature 2", ...],
  "keywords": "keyword1, keyword2, ..."
}}

Be specific about what you see in the image. Mention colors, materials, design elements, and any distinctive features.'

    return prompt

# Test prompt creation
test_prompt = create_product_listing_prompt(
    product_name="Wireless Bluetooth Headphones",
    price=79.99,
    category="Electronics",
    additional_info="Noise cancelling, 30-hour battery"
)

print("\n" + "="*50)
print("PROMPT TEMPLATE")
print("="*50)
print(test_prompt[:500] + "...")
```

[7] ✓ 0.0s Python

```
...
=====
PROMPT TEMPLATE
=====
You are an expert e-commerce copywriter. Analyze the product image and create a compelling product listing.

Product Information:
- Name: Wireless Bluetooth Headphones
- Price: $79.99
- Category: Electronics
- Additional Info: Noise cancelling, 30-hour battery

Please create a professional product listing that includes:

1. **Product Title** (catchy, SEO-friendly, 60 characters max)
2. **Product Description** (detailed, 150-200 words)
   - Highlight key features and benefits
   - Use persuasive ...
```

```
encoded_image = encode_pil_image_to_base64(product_row["image"])

prompt = create_product_listing_prompt(
    product_name=product_row["productDisplayName"],
    price=placeholder_price,
    category=product_row["masterCategory"],
    additional_info=f"{product_row['baseColour']} {product_row['articleType']}, {product_row['usage']} wear"
)

response = client.chat.completions.create(
    model="gpt-4o",
    messages=[
        {
            "role": "user",
            "content": [
                {"type": "text", "text": prompt},
                {
                    "type": "image_url",
                    "image_url": {
                        "url": f"data:image/jpeg;base64,{encoded_image}"
                    }
                }
            ]
        }
    ],
    max_tokens=800
)

content = response.choices[0].message.content

try:
    listing = json.loads(content)
except json.JSONDecodeError:
    listing = {"raw_response": content}

return listing

# Test for the first product
sample_listing = generate_product_listing(products_df.iloc[0])
print(json.dumps(sample_listing, indent=2))
```

[8] ✓ 6.0s Python

```
{
  "raw_response": "`json\n\n  \"title\": \"Stylish Navy Blue Checkered Men's Casual Shirt\", \n  \"description\": \"
```



```
.env  .gitignore U  product_listing_generator.ipynb U  ...
product_listing_generator.ipynb > import time

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api_lab_conda (Python 3.11.15)

try:
    listing = generate_product_listing(product_row)
    listing["product_id"] = int(product_row["id"])
    listing["original_name"] = product_row["productDisplayName"]
    results.append(listing)
    print(f" ✓ Success")

except Exception as e:
    print(f" ⚠ Error: {e}")
    results.append({
        "product_id": int(product_row["id"]),
        "original_name": product_row["productDisplayName"],
        "error": str(e)
    })

# ADD delay between calls to avoid hitting rate limits good practice for batch API calls.
time.sleep(1)

return results

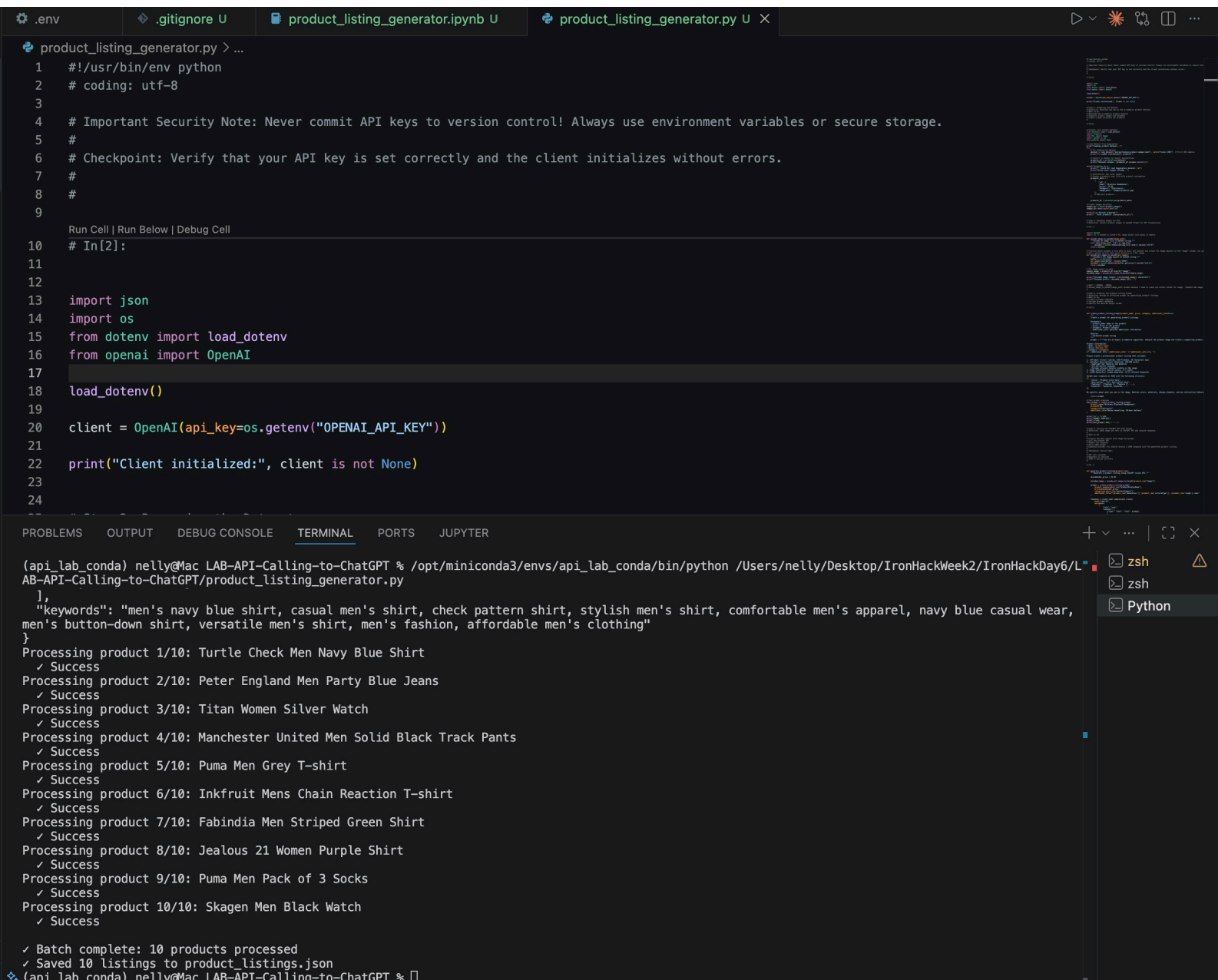
# Run batch for first 10 products
batch_results = process_products_batch(products_df, num_products=10)

print(f"\n✓ Batch complete: {len(batch_results)} products processed")

[11] ✓ 44.7s Python

... Processing product 1/10: Turtle Check Men Navy Blue Shirt
    ✓ Success
Processing product 2/10: Peter England Men Party Blue Jeans
    ✓ Success
Processing product 3/10: Titan Women Silver Watch
    ✓ Success
Processing product 4/10: Manchester United Men Solid Black Track Pants
    ✓ Success
Processing product 5/10: Puma Men Grey T-shirt
    ✓ Success
Processing product 6/10: Inkfruit Mens Chain Reaction T-shirt
    ✓ Success
Processing product 7/10: Fabindia Men Striped Green Shirt
    ✓ Success
Processing product 8/10: Jealous 21 Women Purple Shirt
    ✓ Success
Processing product 9/10: Puma Men Pack of 3 Socks
    ✓ Success
Processing product 10/10: Skagen Men Black Watch
    ✓ Success

✓ Batch complete: 10 products processed
```



Generated Product Listings



Stylish Turtle Check Navy Blue Men's Shirt

Elevate your casual wardrobe with the Turtle Check Men Navy Blue Shirt, a blend of classic style and modern comfort. This shirt features a stylish navy blue and white check pattern that's perfect for any casual outing. Crafted from high-quality, breathable cotton, it ensures all-day comfort while maintaining a crisp, tailored appearance. The shirt's versatile design is ideal for both relaxed and semi-formal occasions. With a classic button-down collar and a front chest pocket, this shirt not only looks great but offers practicality. Perfect for pairing with jeans or chinos, it adapts to any wardrobe seamlessly. Experience the perfect balance of durability and style with this essential wardrobe addition.

- Navy blue and white check pattern
- High-quality, breathable cotton fabric
- Classic button-down collar
- Front chest pocket
- Versatile for casual and semi-formal wear
- Tailored fit for enhanced comfort
- Durable and easy to maintain

Keywords: men's shirts, navy blue shirt, check pattern shirt, casual wear, button-down shirt, cotton shirt, Turtle Check shirt, men's fashion, stylish men's shirt, breathable fabric, classic design, versatile shirt, men's apparel, fashion essentials, wardrobe basics



Stylish Peter England Blue Party Jeans for Men

Elevate your party style with Peter England Men's Party Blue Jeans. These trendy blue jeans are perfect for any casual or party setting, offering a sleek and modern look that complements any outfit. Crafted from high-quality denim, these jeans provide both comfort and durability, ensuring you move with ease throughout the day or night. The deep blue fabric features subtle fading and whiskering for a contemporary touch, while the tailored fit enhances your silhouette. Whether paired with a crisp shirt or a casual tee, these jeans are designed to be your go-to choice for versatility and trendsetting style. Experience the perfect blend of fashion and function with Peter England, a brand synonymous with quality and sophistication.

- High-quality denim material
- Deep blue color with stylish fading
- Comfortable and durable fit
- Perfect for casual and party occasions
- Contemporary whiskering details
- Tailored fit to enhance silhouette
- Versatile styling options

Keywords: Peter England, men's jeans, blue jeans, party jeans, casual wear, durable denim, stylish men's clothing, trendy jeans, casual jeans, tailored fit, whiskering jeans, deep blue jeans



Elegant Titan Silver Women's Watch - Chic Casual Style

Elevate your everyday style with the Titan Women Silver Watch, a perfect blend of elegance and modern design. Crafted with a sleek silver stainless steel band, this watch exudes sophistication while providing comfort for all-day wear. The minimalist silver dial is accentuated by bold hour markers and elegant watch hands, making it easy to read at a glance. Ideal for casual outings or professional settings, this watch seamlessly complements any outfit. Durable construction ensures it withstands the rigors of daily use, while its timeless design keeps you fashionable year-round. Whether gifting or treating yourself, the Titan Silver Watch is a statement piece that embodies grace and charm.

- Sleek silver stainless steel band
- Minimalist silver dial with bold markers
- Comfortable and durable for daily wear
- Timeless and versatile design
- Perfect for casual and professional settings
- Reliable timekeeping
- Ideal gift option

Keywords: Titan watch, silver watch, women's watch, casual watch, elegant watch, stainless steel watch, minimalist watch, fashion accessory, women's accessories, timeless design, durable watch, stylish wristwatch, affordable watch, chic style